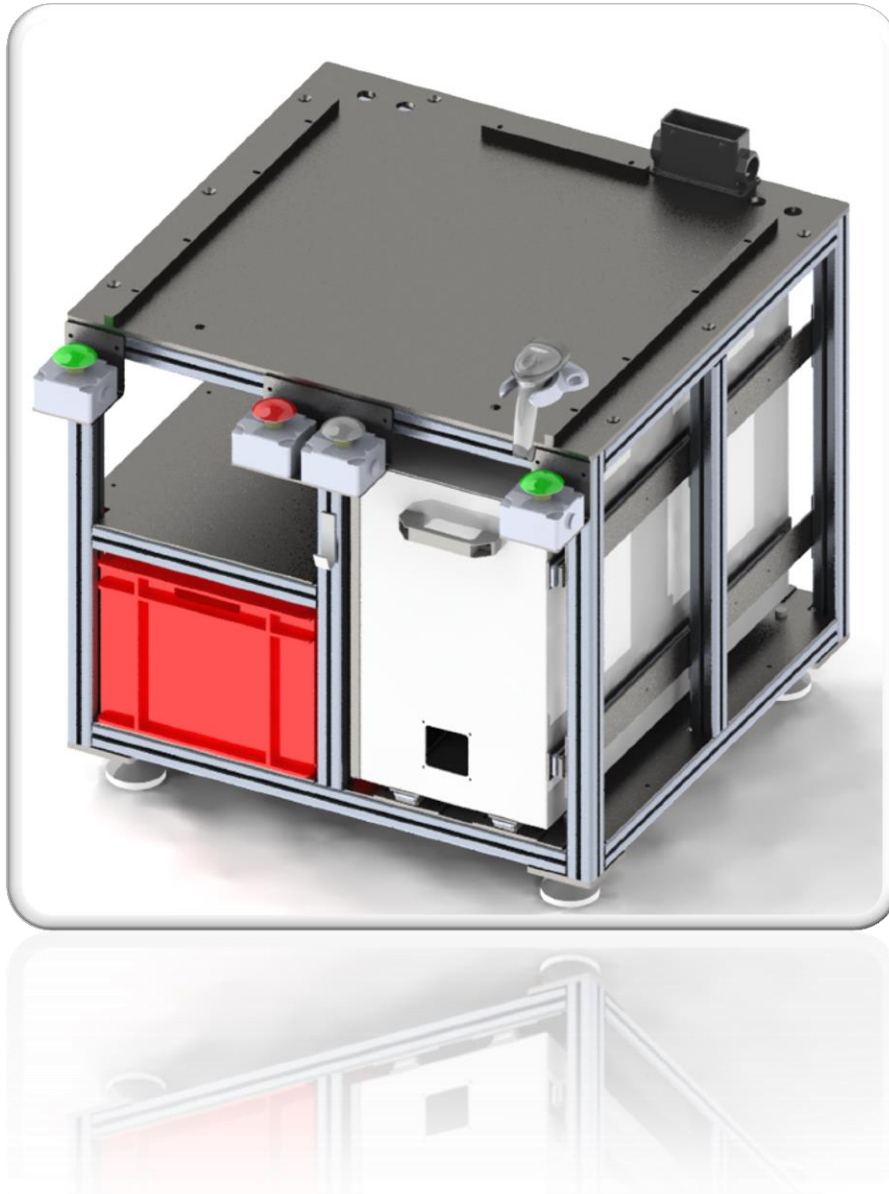


USER MANUAL

STATION 1: HOUSING TO REFLECTOR ASSEMBLY-LH STATION 2: HOUSING TO LIGHT GUIDE CARRIER ASSEMBLY-LH



CLIENT NAME:

MARELLI MOTHERSON AUTOMOTIVE LIGHTING INDIA PRIVATE LIMITED
Ambethan 165/1

MACHINE MANUFACTURED BY:

INFINITY AUTOMATION SYSTEMS PVT LTD.

Unit 2:S. No. 80/2A, Mumbai Bangalore Bypass Highway

Near JSPM Institute,

Tathawade, Pune-411033.

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1. INTRODUCTION

We are certain that you have made the right choice with our product and thank for your trust.

For your personal safety

Before carrying out any activities on or with the PCBA+Metal Plate +EMI Shield Assembly LH & RH Station, please carefully read these operating instructions first, in particular the chapter on "Safety regulations".

Application area of these operating instructions

These operating instructions describe the PCBA+Metal Plate +EMI Shield Assembly LH & RH Station and provide all information required for safe operation and maintenance of functionality. Please keep these operation instructions available at all times for all persons involved with the machine.

Questions or unclear points

Please contact immediately if you have any questions or unclear points:
Original language of assembly instructions is English.

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2. SAFETY REGULATIONS

2.1 Term definitions

The following terms are used with regards to safety:

Personnel qualification

The work to be implemented with or on the product is described in the corresponding chapters in the operating instructions. The personnel qualifications listed in this chapter (in "Introduction") describes the minimum requirements for personnel regarding training and competences for implementing this work.

Personnel

Personnel are people who meet the personnel qualifications. Only such personnel are authorized to implement the work described in the applicable chapters.

User

Users are persons who cannot be identified as personnel but who still carry out activities with or on the product.

Operator

The operator is the owner of the product. This is irrespective of whether the product is:

- Operated as a stand-alone machine
- Operated as part of a system or
- Operated by a third party

Product life phases

The product life phases are the state or usage phases of the product. These range from leaving the assembly hall to disassembly and disposal, and are divided into correspondingly logical sections (chapters) in these operating instructions.

2.2 Significance of the safety regulations

These safety regulations contain all behavior regulations and generally applicable danger information. The safety of the user and the personnel can only be ensured when these safety regulations and the safety information listed in the individual chapters are strictly complied with and followed. The operator is responsible for the enforcement.

2.3 Information and symbols

2.3.1 Warning and safety notices

The warning and safety notices used in these operating instructions have the following significance:

	 CAUTION Indicates a hazard with low risk which, if not avoided, could result in minor or moderate injury.
--	--

	 WARNING Indicates a possible hazard with medium risk which, if not avoided, could result in death or serious injury.
--	---

	 DANGER Indicates an immediate hazard with high risk which, if not avoided, will result in death or serious injury.
--	---

NOTICE	
Indicates practical information and tips that permit the best use of machinery, system or equipment.	

CAUTION
To warn of material damage.

2.3.2 Notes and symbols in the manual.

The following information and symbols are used in manual.

Warning signs	Mandatory sign	Mandatory sign
Danger of hand injuries	Wear safety goggles	Wear gloves
		

2.4 Application limitations

The following technical operating instructions and ambient conditions must be complied with for the operation of the PCBA+Metal Plate +EMI Shield Assembly LH & RH Station.

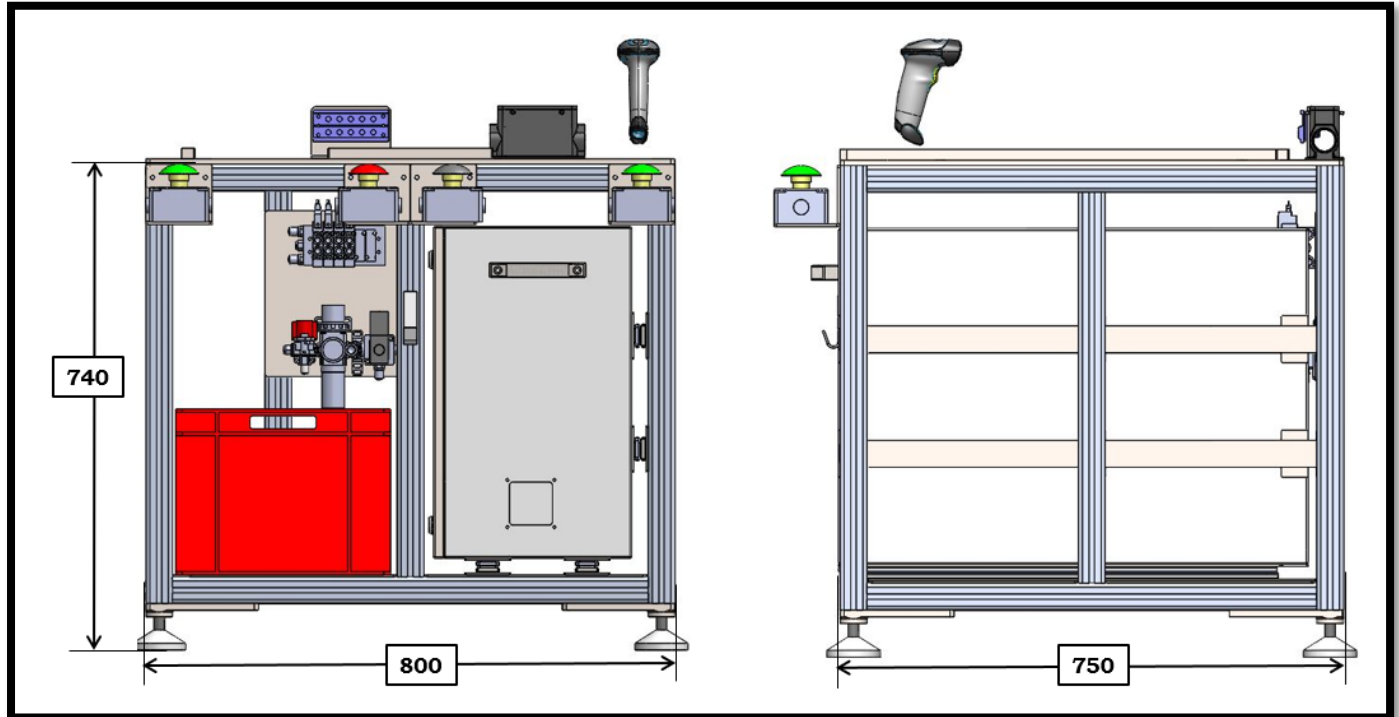
Environmental conditions	
Working temperature	+10 ... +50 °C
Storage temperature	- 10 ... +70 °C
Relative humidity	Max. 90%, non-condensing
Mechanical power supply	
Operating pressure	2-6 bar
Mechanical	
Machine Weight	150kg Approx.
Electrical power supply	
Supply voltage	230 VAC
Input Power	8 kW
Full Load Current	16 Amp
Primary Fuse	32 Amp
Max. Short Circuit Interrupting Cap	14 Amp
Control Voltage	24 V

2.5 Do's and Don'ts for PCBA+Metal Plate +EMI Shield Assembly LH & RH Station

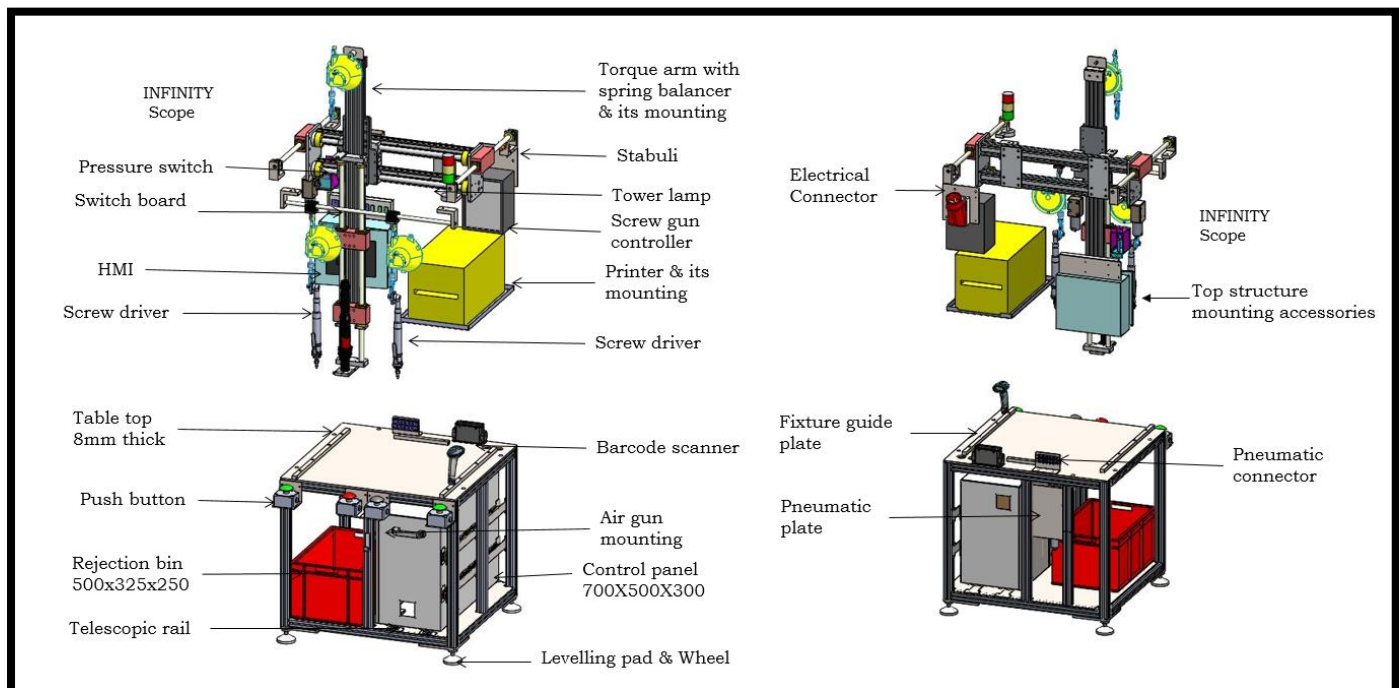
- Regular checking of penumatics parts.
- Regular checking of drag chain.
- Regular checking 2/2 valve.

3. DESCRIPTION OF THE MACHINE

3.1 Layout of Station:



3.2 Description of Station:



3.3 Assembly Drawings:

**** IF ANY DOUBT PLEASE ASK ****

NOTES:-

1. **WHETHER WELD, USE 2 DOWEL & 2 BOLT BEFORE WELDING.**
2. **DO NOT WELD ALUMINIUM MATERIAL WITHOUT SPECIAL NOTE.**
3. **PART SHALL BE FREE FROM ALL KINDS OF DEFECTS.**
4. **UNWANTED CHAMFER SHALL BE 18-45° ALL OVER.**
5. **UNWANTED BURR AFTER MACHINING.**
6. **UNWANTED TOLERANCE FOR SLOT OR HOLE.**
- 6.1 **POSITION TOLERANCE FOR DOWEL HOLE WILL BE WITHIN ±0.1 mm.**
- 6.2 **POSITION TOLERANCE FOR DOWEL HOLE WILL BE WITHIN ±0.2 mm.**
- 6.3 **WHEREVER PIN / SHAFT (0.1 mm).**

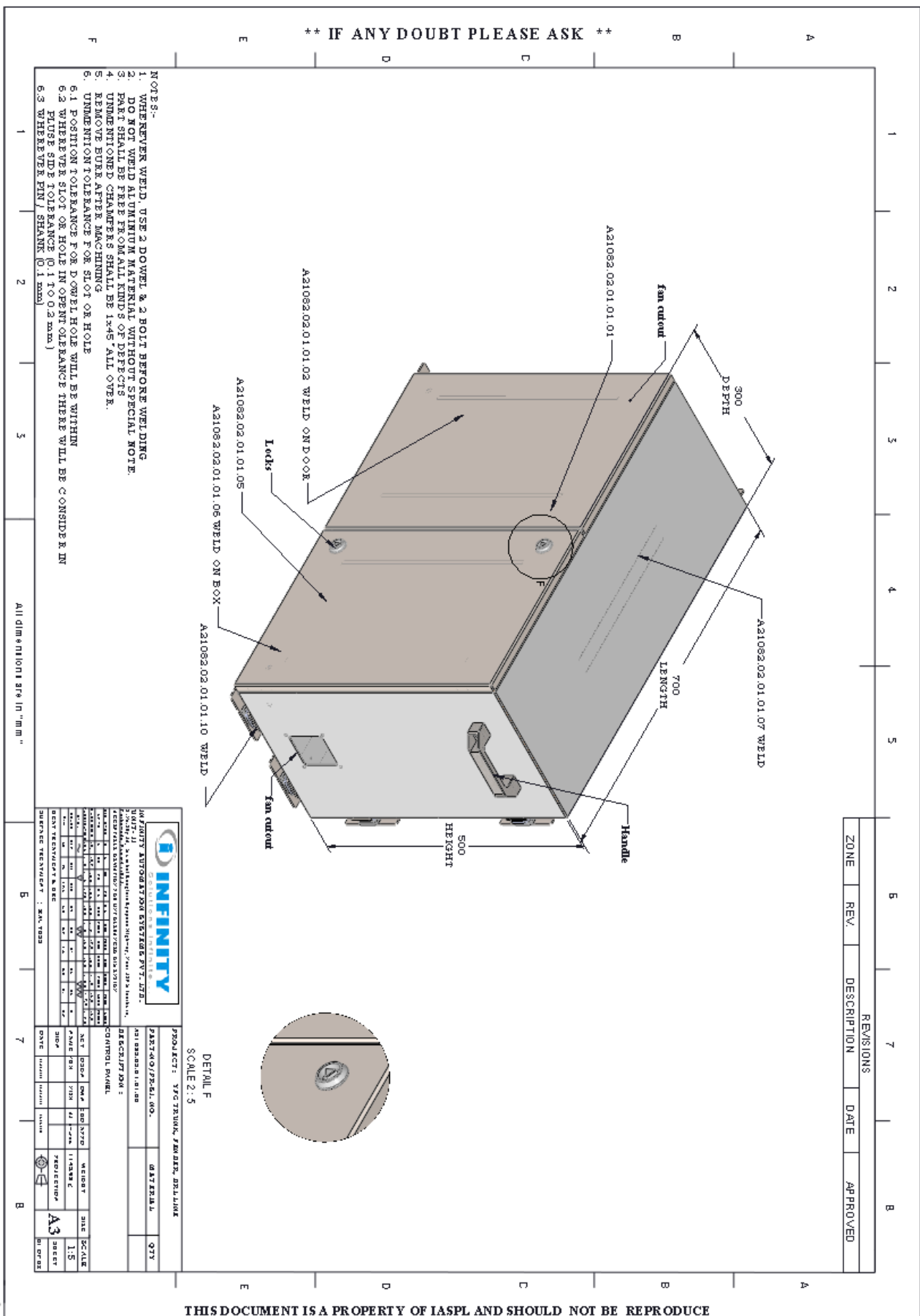
USE BLUE COLOR FOR FORWARD AND BLACK COLOR FOR BACKWARD PNEUMATIC PIPE.

ALL dimensions are in "mm"

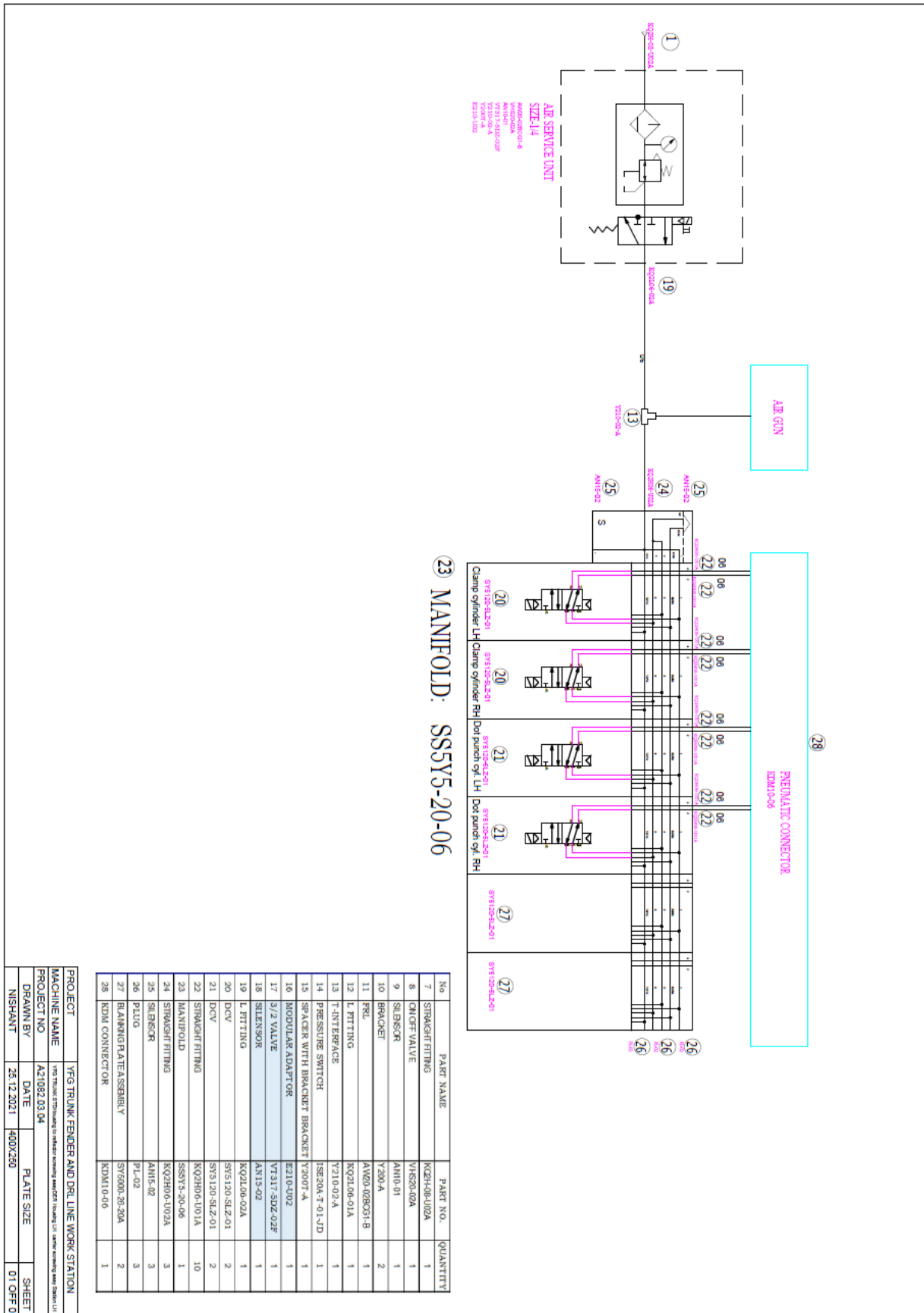
REVISIONS			
ZONE	REV.	DATE	APPROVED
1	1	2020.03.01.01.00	CONTROL PANEL
2	2	2020.03.01.01.01	PLATE
3	3	2020.03.01.01.02	SEPARATOR PLATE
4	4	2020.03.01.01.05	GUIDE PLATE
5	5	2020.03.01.01.04	GUIDE PLATE 2
6	6	2020.03.01.01.05	GUIDE PLATE 2
7	7	2020.03.01.01.07	WHEEL MOUNTING PLATE
8	8	2020.02.08.01.04	WHEEL
9	9	2020.02.08.01.04	WHEEL
10	10	2020.02.08.01.04	WHEEL
11	11	2020.02.08.01.04	WHEEL
12	12	2020.02.08.01.04	WHEEL
13	13	2020.02.08.01.04	WHEEL
14	14	2020.02.08.01.04	WHEEL
15	15	2020.02.08.01.04	WHEEL
16	16	2020.02.08.01.04	WHEEL
17	17	2020.02.08.01.04	WHEEL
18	18	2020.02.08.01.04	WHEEL
19	19	2020.02.08.01.04	WHEEL
20	20	2020.02.08.01.04	WHEEL
21	21	2020.02.08.01.04	WHEEL
22	22	2020.02.08.01.04	WHEEL
23	23	2020.02.08.01.04	WHEEL
24	24	2020.02.08.01.04	WHEEL
25	25	2020.02.08.01.04	WHEEL
26	26	2020.02.08.01.04	WHEEL
27	27	2020.02.08.01.04	WHEEL
28	28	2020.02.08.01.04	WHEEL
29	29	2020.02.08.01.04	WHEEL
30	30	2020.02.08.01.04	WHEEL
31	31	2020.02.08.01.04	WHEEL
32	32	2020.02.08.01.04	WHEEL
33	33	2020.02.08.01.04	WHEEL

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	A21082.03.01.01.00	CONTROL PANEL	1
2	A21082.03.01.01.01	PLATE	1
3	A21082.03.01.01.02	SEPARATOR PLATE	1
4	A21082.03.01.01.05	GUIDE PLATE	1
5	A21082.03.01.01.04	GUIDE PLATE 2	1
6	A21082.03.01.01.05	GUIDE PLATE 2	1
7	A21082.03.01.01.07	WHEEL MOUNTING PLATE	1
8	A21082.02.08.01.04	WHEEL	2
9	A21082.02.08.01.04	WHEEL	2
10	A21082.02.08.01.04	WHEEL	2
11	A21082.02.08.01.04	WHEEL	2
12	A21082.02.08.01.04	WHEEL	2
13	A21082.02.08.01.04	WHEEL	2
14	A21082.02.08.01.04	WHEEL	2
15	A21082.02.08.01.04	WHEEL	2
16	A21082.02.08.01.04	WHEEL	2
17	A21082.02.08.01.04	WHEEL	2
18	A21082.02.08.01.04	WHEEL	2
19	A21082.02.08.01.04	WHEEL	2
20	A21082.02.08.01.04	WHEEL	2
21	A21082.02.08.01.04	WHEEL	2
22	A21082.02.08.01.04	WHEEL	2
23	A21082.02.08.01.04	WHEEL	2
24	A21082.02.08.01.04	WHEEL	2
25	A21082.02.08.01.04	WHEEL	2
26	A21082.02.08.01.04	WHEEL	2
27	A21082.02.08.01.04	WHEEL	2
28	A21082.02.08.01.04	WHEEL	2
29	A21082.02.08.01.04	WHEEL	2
30	A21082.02.08.01.04	WHEEL	2
31	A21082.02.08.01.04	WHEEL	2
32	A21082.02.08.01.04	WHEEL	2
33	A21082.02.08.01.04	WHEEL	2

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3.4 Pneumatic Drawings:





Screw Gun



Barcode scanner & Printer Mounting

3.5 Operation

3.5.1 Safety regulations for PCBA +Metal Plate +EMI Shield Assembly LH & RH Station

1. Any type of maintenance work done by authorized person.
2. Always turn off Machine switch while any type of maintenance work.

3.5.1.1 Personal Safety Equipment

All users are obliged to wear the following personal safety equipment for all work:

- Tightly fitting working clothes
- Safety goggles
- Gloves

3.5.2 Steps for Operation Sequence:

- Pick LH housing from bin.
- Do the barcode scan of wire harness for variant confirmation
- Connect LH side connector for glow test.
- Place on fixture (Lens side up)
- Pick reflector sub assembly.
- Do the PCB barcode scan. (PCB barcode data & Bin number to be confirm with master data)
- Do the LH PCB lumber connection with LH wire harness connector.
- Locate reflector on housing.
- Press cycle start.(Part gets clamped – If stud bolt & stud pin presence check OK)
- PCB glow test light guide checking start. (if PCB glow test (current check) ok then screwdriver enable)
- Pick screwdriver.
- Pick screw (3 no's).
- Start screwing LH carrier to LH housing (3nos. on LH part.)
- If screwing ok then dot punch operate
- Then barcode generate (PCB scanned barcode detail as it is) & de-clamp to part
- Remove barcode sticker from printer.
- Paste on housing LH.
- Remove LH side glow test connector.
- Unload LH housing from fixture.

4. LIFTING, TRANSPORTATION & PACKAGING:

4.1 Lifting Arrangement:

For lifting of machine, there is fork lift.

4.2 Transportation:

Transfer the final test workstation and all of its separately packed associated components, in a closed vehicle or closed shipping container only. Each component of this machine should remain horizontal at all times during transport.

A fork truck should be used to load/offload the pallet holding this machine into/from vehicles.

4.3 Packaging:

Put final test workstation at an appropriate transport pallet that corresponds to the weight, and secure it against sliding. Protect each table from dirt and moisture by foil. Then house it completely.

Remove all components and subassemblies that are not permanently attached to the final test workstation, wrap them in plastic foil, and pack them in separate shipping containers.

Secure Component groups at the final test workstation against uncontrolled movement.

5. HMI SCREEN:

5.1. MAIN SCREEN:



FIG.1.MAIN SCREEN

➤ The software has menu buttons as given below:

1. Program
2. Auto
3. Manual
4. Input
5. Output
6. Configuration
7. Alarm History
8. Data log Screen

5.2. PROGRAM SCREEN:

- Click on '**PROGRAM**' button, below screen appears.

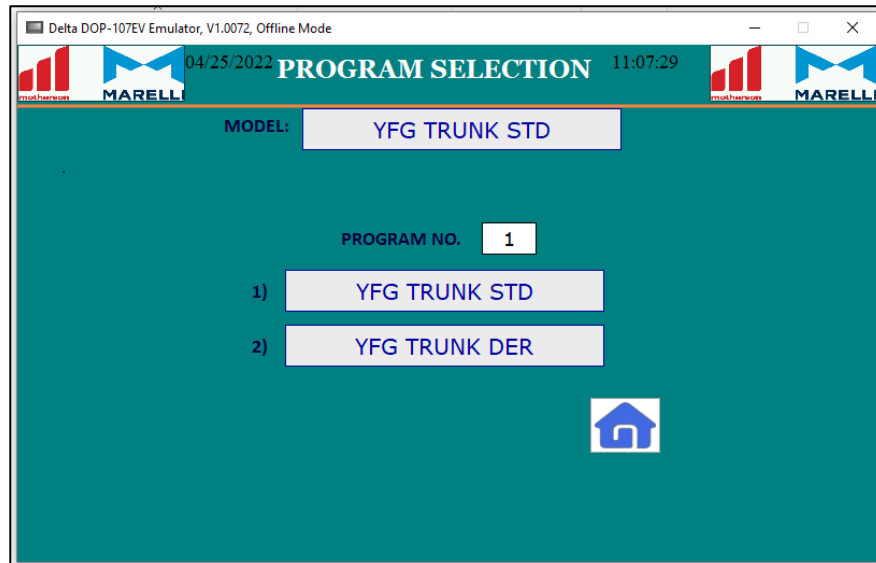


FIG.2.PROGRAM SCREEN

- Here for program selection enter one or two in Program No. field
- Description is displayed in the fields below Program No.

5.3. AUTO:

- Click on 'AUTO' button, below screen appears.

The screenshot displays the 'AUTO' screen with the following elements:

- Header:** Includes logos for 'MARELL' and 'INFINITY', a date field 'mm/dd/yy', the title 'AUTO', and a time field 'HH:MM:SS'.
- Model and Reset:** A 'MODEL:' field containing 'ABCDEFGHIJKLMNO' and a 'RESET' button.
- Buttons:** 'RESET COUNTER', 'FIXTURE NOK', 'WH SCAN', and 'PCB SCAN'.
- Barcode Area:** A large field for a barcode, currently showing 'ABCDEFGHIJKLMNOPQRSTUVWXYZABCDEFGHI'.
- Studs and Pins:** 'STUD BOULT 1', 'STUD BOULT 2', and 'STUD PIN' each with a red indicator light.
- GLOW TEST Section:**
 - 'BIN No:' field with 'ABCD'.
 - Table for lamp tests:

	MIN	ACTUAL	MAX
STOP LAMP	123	123 mA	123
TAIL LAMP	123	123 mA	123
- Counts and Status:** 'OK COUNT' (12345), 'CYCLE TIME' (1234), 'NOK COUNT' (1234), 'Screw Gun Enable' button, and 'SCREW GUN COUNT' (1234).
- Message Area:** A field labeled 'Message' with '####'.

FIG.3.AUTO

- This screen display live cycle status.
- This screen shows cycle time in second.
- Here also we display sensors, if it is present then it indicates with Green color and if it is not then indicates with Red color.
- It also indicates OK and NOK part count & screw count.
- Glow test values of each lamp are displayed with their Min & Max current ranges along with Bin No.

5.4. MANUAL:

- Click on 'MANUAL' button, below screen will appear.

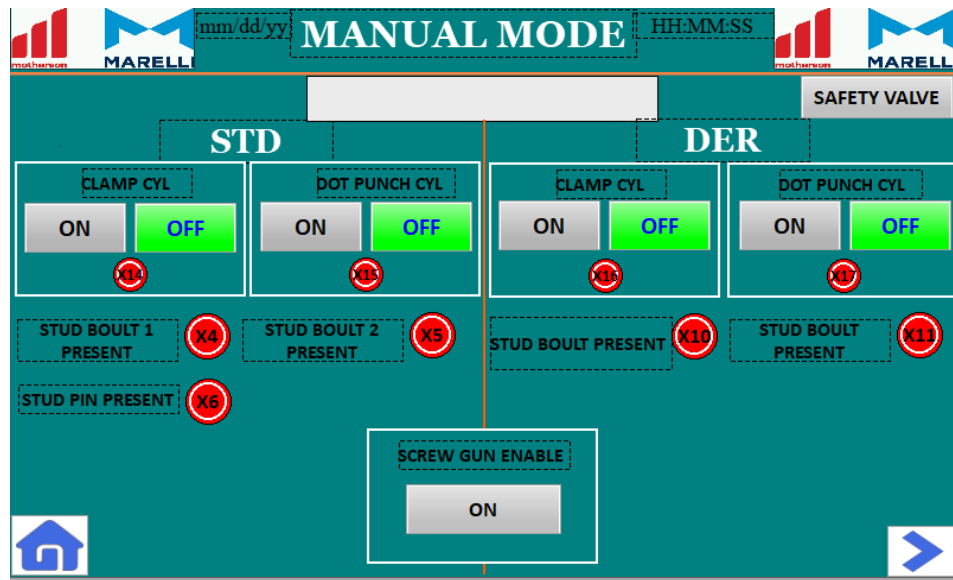


FIG.4.1.MANUAL OPERATION

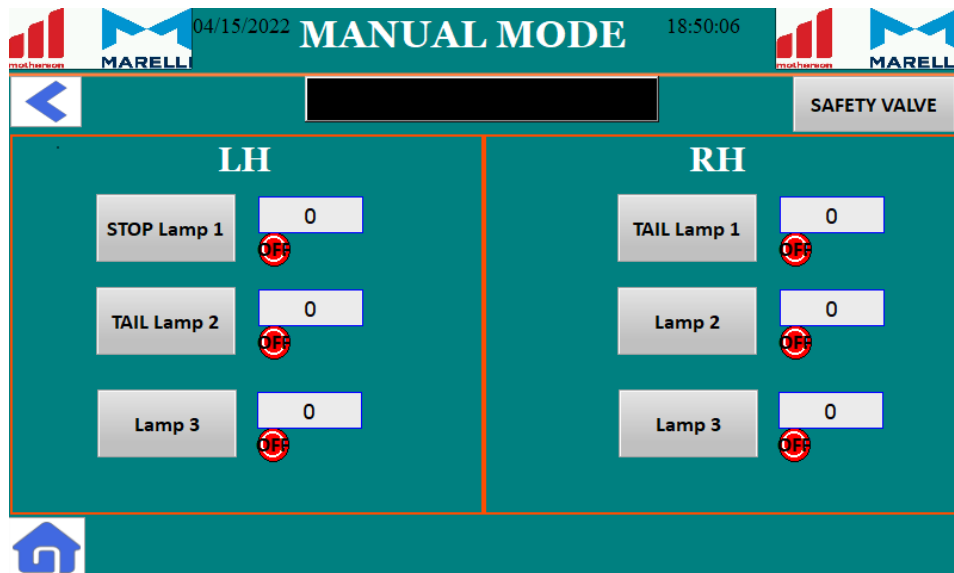


FIG.4.2.MANUAL OPERATION

- Here all the cylinders can be operated manually after switching the safety valve ON.
- Screw gun can be enable & disable manually.
- Press **HOME** button to go back.
- Here we can manually turn on the lamp and the value of current display on the screen.

5.5. INPUT LIST:

- Click on 'INPUT' button, below screen will appear.

INPUTS	
Cycle Start PB 1	X0
Cycle Start PB 2	X1
Emergency PB	X2
Pressure Switch	X3
STD Stud Bolt Present Sensor 1	X4
STD Stud Bolt Present Sensor 2	X5
STD Stud Pin Present Sensor	X6
STD Part Present 4(Spare)	X7
DER Stud Bolt Present Sensor1	X10
DER Stud Bolt Present Sensor2	X11
DER Part Present 3 (Spare)	X12
DER Part Present 4 (Spare)	X13
STD Clamp Cyl 1 Fwd RS	X14
STD Dot Punch Cyl Fwd RS	X15
DER Clamp Cyl 1 Fwd RS	X16
DER Dot Punch Cyl Fwd RS	X17

FIG.5.1.INPUT LIST

INPUTS	
LH/RH Fix Sel 1	X20
Ip Spare	X21
Screw Gun Nok	X22
Screw Gun OK	X23
LH/RH Fix Sel 2	X24
Ip Spare	X25
Ip Spare	X26
Ip Spare	X27
Ip Spare	X30
Ip Spare	X31
Ip Spare	X32
Ip Spare	X33
Ip Spare	X34
Ip Spare	X35
Ip Spare	X36
Ip Spare	X37

FIG.5.2.INPUT LIST

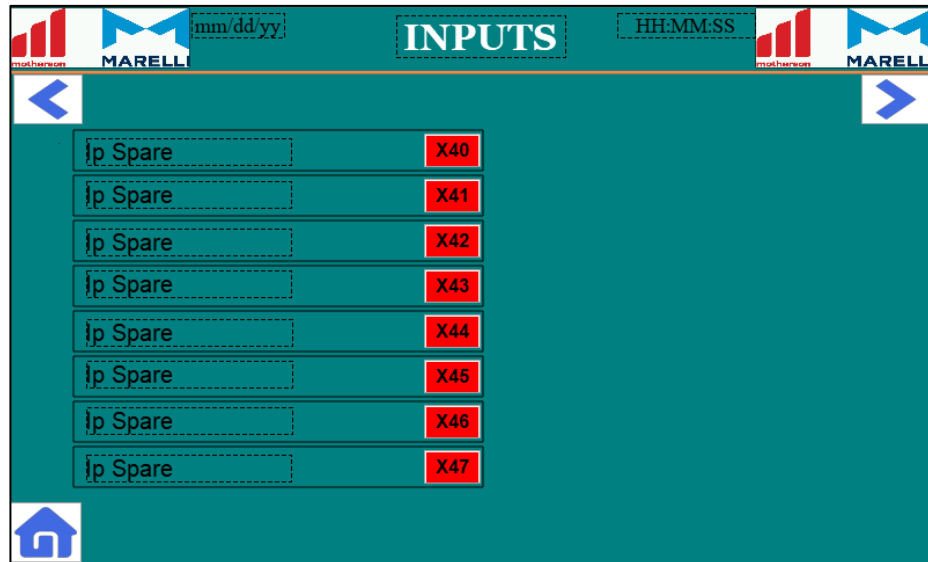
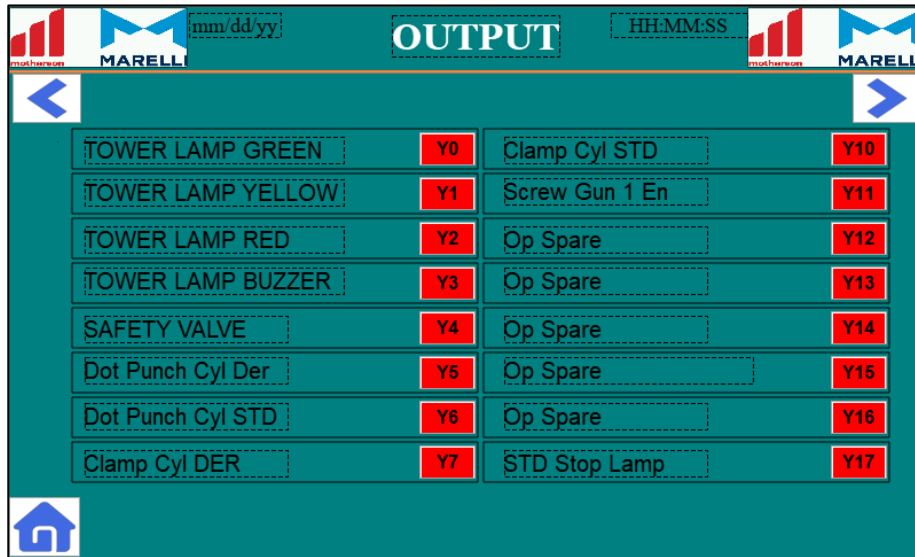


FIG.5.3.INPUT LIST

- For maintenance purpose all the inputs are displayed on the above screens.
- If input is ON, indicator turns GREEN. If it is OFF, indicator turns RED.
- Press forward arrow button for go to next window.
- Press backward arrow button for go back.
- Press HOME button to go on main screen.

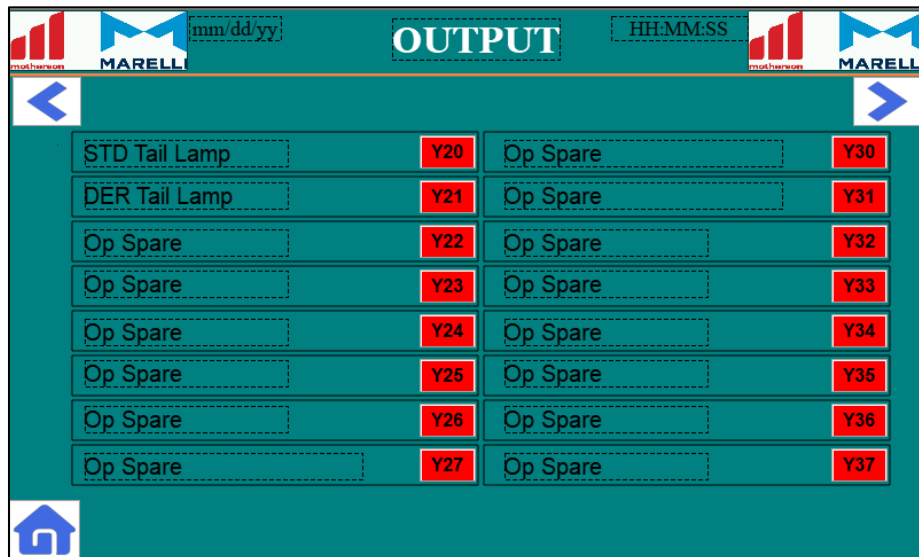
5.6. OUTPUT LIST:

➤ Click on 'OUTPUT' button, below screen will appear.



Output	Function
Y0	TOWER LAMP GREEN
Y1	TOWER LAMP YELLOW
Y2	TOWER LAMP RED
Y3	TOWER LAMP BUZZER
Y4	SAFETY VALVE
Y5	Dot Punch Cyl Der
Y6	Dot Punch Cyl STD
Y7	Clamp Cyl DER
Y10	Clamp Cyl STD
Y11	Screw Gun 1 En
Y12	Op Spare
Y13	Op Spare
Y14	Op Spare
Y15	Op Spare
Y16	Op Spare
Y17	STD Stop Lamp

FIG.6.1. OUTPUT LIST



Output	Function
Y20	STD Tail Lamp
Y21	DER Tail Lamp
Y22	Op Spare
Y23	Op Spare
Y24	Op Spare
Y25	Op Spare
Y26	Op Spare
Y27	Op Spare
Y30	Op Spare
Y31	Op Spare
Y32	Op Spare
Y33	Op Spare
Y34	Op Spare
Y35	Op Spare
Y36	Op Spare
Y37	Op Spare

FIG.6.2. OUTPUT LIST

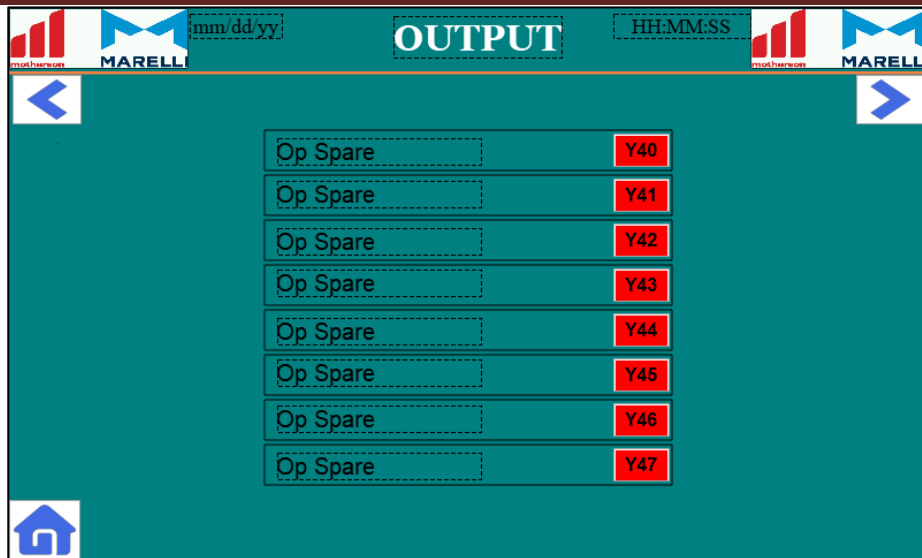


FIG.6.3. OUTPUT LIST

- For maintenance purpose all the outputs are displayed on the below screens.
- If output is ON then indicator turns GREEN if it is OFF then indicator turns RED.
- Press forward arrow button for go to next window.
- Press backward arrow button for go back.
- Press HOME button to go on main screen.

5.7. CONFIGURATION:

- Click on 'CONFIGURATION' button, below screen will appear.

The screenshot shows the 'CONFIGURE' screen with a teal background. At the top, there's a header bar with 'mm/dd/yy' and 'HH:MM:SS' fields, and a 'MARELLI' logo. Below the header, the 'PROG NO.' is set to '12' and the 'SCREW COUNT' is '1234'. A large blue arrow points right. The main area contains several test parameters, each with a red 'X' icon indicating a status or error:

- CLAMP CYL 1
- DOT PUNCH CYL
- STUD BOULT 1 PRESENSE
- STUD BOULT 2 PRESENSE
- CLAMP CYL 1 REED SW
- DOT PUNCH REED SW
- STUD PIN PRESENSE
- PRE GLOW TEST
- POST GLOW TEST

At the bottom left is a home icon.

FIG.7.1.CONFIGURATION

The screenshot shows the 'CONFIGURE' screen with a teal background. At the top, there's a header bar with 'mm/dd/yy' and 'HH:MM:SS' fields, and a 'MARELLI' logo. Below the header, the 'PROG NO.' is set to '12' and the 'SCREW COUNT' is '1234'. A large blue arrow points right. The main area contains several test parameters, each with a red 'X' icon indicating a status or error:

- FIXTURE ID
- CURRENT = SET
- 1234 (NOK) 1234
- LAMP GLOW TEST SELECTION
- STOP LAMP
- TAIL LAMP
- BIN WISE CURRENT
- GLOW COUNTS 12345 N
- GLOW TIME 12345 Sec

At the bottom left is a home icon.

FIG.7.2. CONFIGURATION

CONFIGURE

mm/dd/yy HH:MM:SS

Without Bin Wise Current

WH SCAN X PCB SCAN X

WIRE HARNESS No. ABCDEFGHIJKL PCB PART No. ABCDEFGHIJKLMN

BIN 7E		BIN 8E		BIN 5F	
	MIN	MAX		MIN	MAX
STOP LAMP	123.4 mA	123.4 mA	STOP LAMP	123.4 mA	123.4 mA
TAIL LAMP	123.4 mA	123.4 mA	TAIL LAMP	123.4 mA	123.4 mA

BIN 6F		
	MIN	MAX
STOP LAMP	123.4 mA	123.4 mA
TAIL LAMP	123.4 mA	123.4 mA

FIG.7.3. CONFIGURATION

CONFIGURE

04/15/2022 18:49:06

YFG TRUNK STD

WITHOUT CURRENT SETTING

	MIN		MAX
STOP LAMP 1	0.2 Amp		0.3 Amp
TAIL LAMP 2	0.1 Amp		0.1 Amp

FIG.7.4. CONFIGURATION

- To open this screen first we have to enter password.
- Here we can create a new program.
- Here we can enable or disable sensors and cylinder by pressing the button.
- Also we can enter screw count and REED SW delay timing.
- From Fig.7.1 we can Enable/Disable each Lamp for Glow Testing, also Select correct fixture ID.
- From Fig7.3 we can input PCB & Wire Harness Part No., Bin wise Glow Test Ranges for each Lamp, Enable/Disable for Scanning.
- If Scanning is bypassed, we have to Input Glow Test Current Ranges in 'Without Current Setting' Screen.
- From Fig.7.4 we can enter Min/Max value for lamp.
- Press **Home** to go main menu.

5.8. ALARM SCREEN:

- Click on ‘**ALARM HISTORY**’ button, below screen will appear.

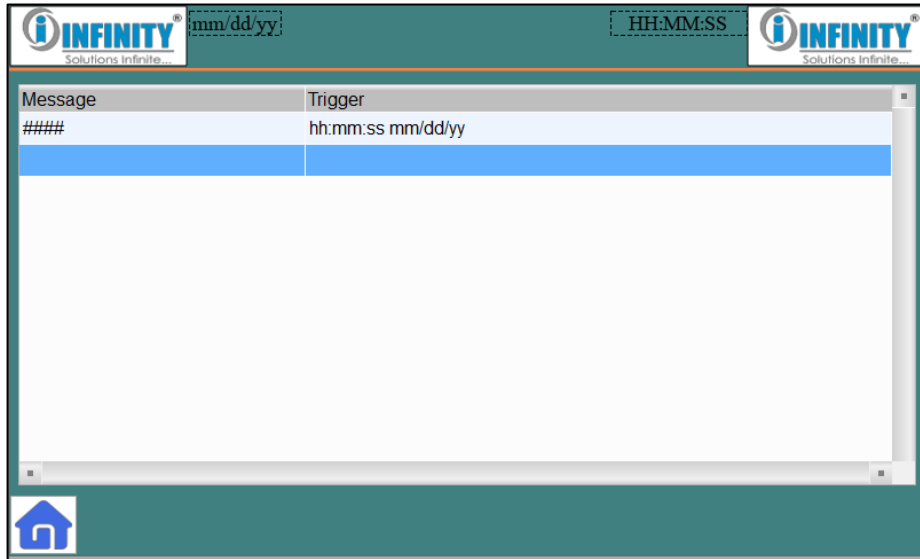


FIG.8. ALARM HISTORY

- Alarm history can be shown on the screen.

5.9. DATALOG SCREEN:

- Click on 'DATA LOG SCREEN' button, below screen will appear.

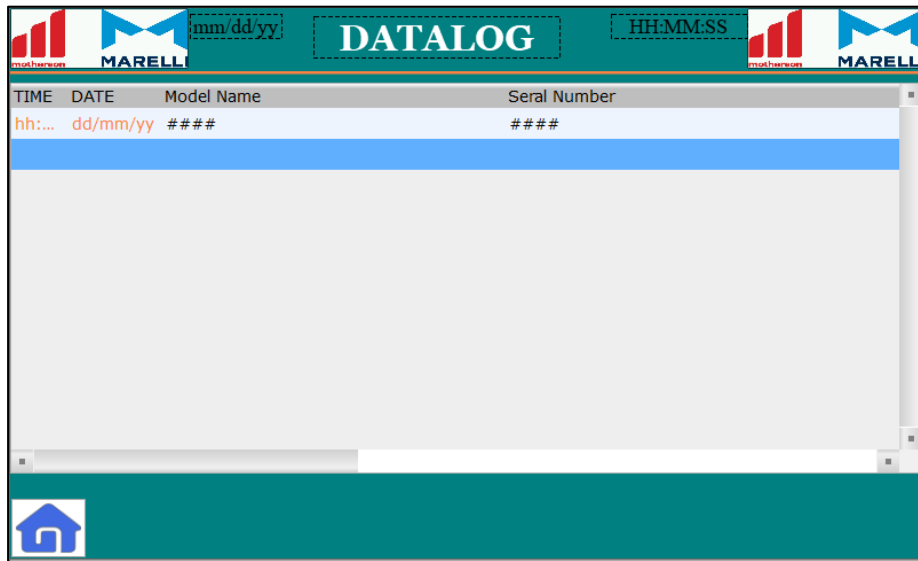


FIG.9. DATALOG SCREEN

- This screen displays the data history of Glow test of previous cycles.

6. SPARE LIST:

• ELECTRICAL BOM:

Project No.: A21082				
NO.	Part Number	Name/Description	Qty	Make
1	DOP-107EV	HMI.HMI 7" In build Ethernet	1	DELTA
2	FX5U-80MT/ESS	PLC.CPU 40 I/P(Sink/Source).40 O/P(Source) 100 to 240 VAC Supply	1	MITSUBHISHI
3	FX5-4AD-ADP	FX5U SERIES ANALOG INPUT 12BIT.	1	MITSUBHISHI
4	6GK5005-0BA00-1AB2	6GK5005-0BA00-1AB2.SCALANCE XB005 unmanaged Industrial Ethernet Switch for 10/100	1	SIEMENS
5	XVGB3S	Tower lamp RAG with Buzzer 24V DC	1	SCHNEIDER
6	ONE WAY SIZE 90X80X52 MM	PENDANT	4	TEKNIC
7	2AM2	PUSH BUTTON. METALLIC BLACK MUSHROOM TYPE PUSHBUTTON, 40MM, MOMENTARY (SPRING RETURN)	1	TEKNIC
8	2AMJ3	PUSH BUTTON. METALLIC GREEN MUSHROOM TYPE PUSHBUTTON, 60MM, MOMENTARY	2	TEKNIC
9	2AML4	PUSH BUTTON. METALLIC RED MUSHROOM TYPE PUSHBUTTON, 40MM, LATCHING (ENERGENCY PUSHBUTTON)	1	TEKNIC
10	S2	NC CONTACT METALLIC RED NORMALLY CLOSED CONTACT ELEMENTS	1	TEKNIC
11	S1	NO CONTACT METALLIC GREEN NORMALLY OPEN CONTACT ELEMENTS	3	TEKNIC
12	M8	QUICK RELEASE OPENING	1	LOCAL
13	555234	INDUSTRIAL PLUG.Straight plug P17 - IP44 - 200/250 V~ - 32 A - 2P+E	1	LEGRAND

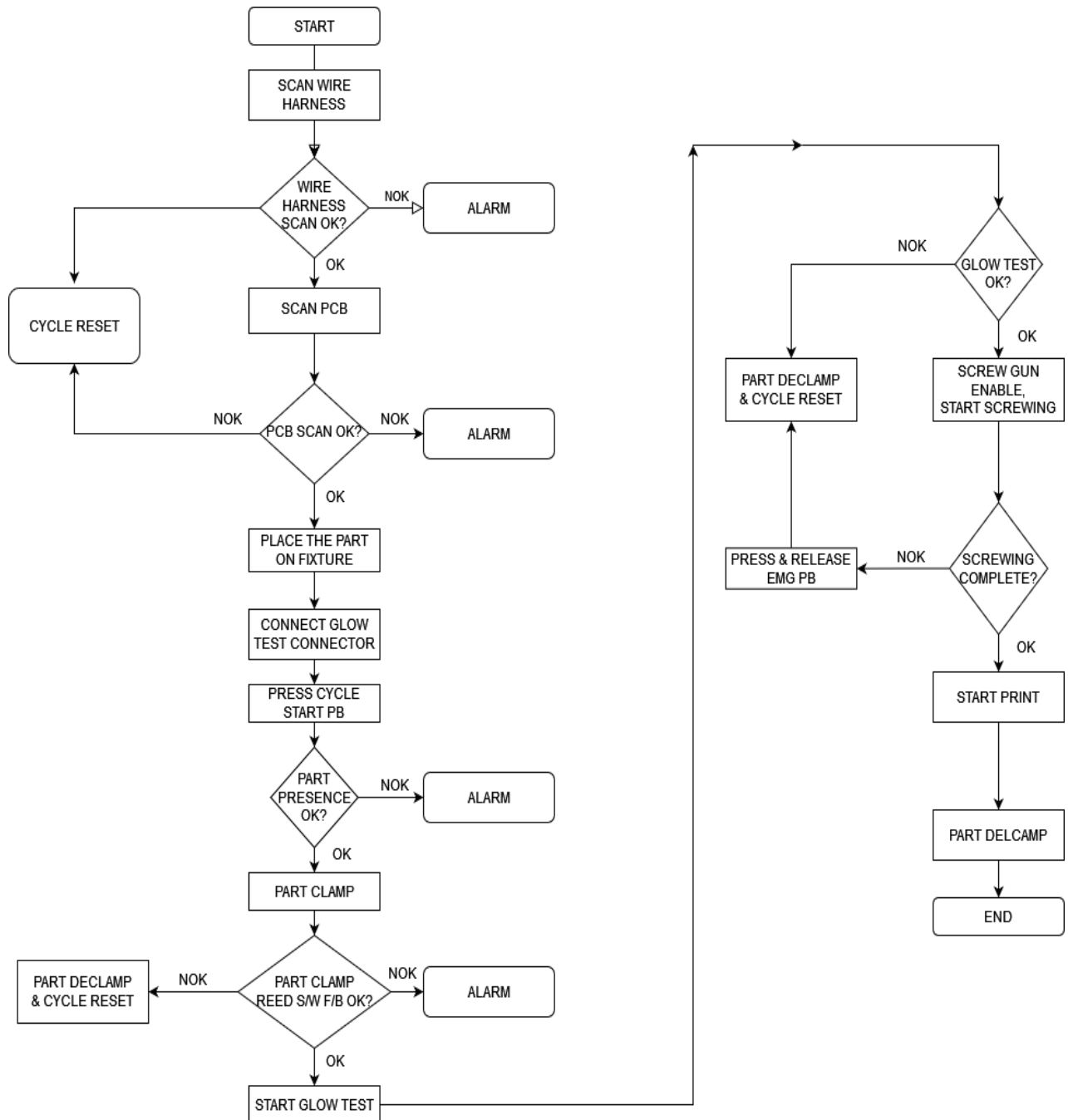
14	555254	INDUSTRIAL SOCKET.Surface mounting socket P17 - IP44 - 200/250 V~ - 32 A - 2P+E	1	LEGRAND
15				
16	6755 69	SERVICE SOCKET - MYLINC PLATES - 9 MODULES	1	LEGRAND
17	GOLDEN-30W-T8 LED(2ft) 6500K DAY-LIGHT	GOLDEN-30W-T8 LED(2ft) 6500K DAY-LIGHT	2	OREVA
18	PD9531	Scanner.Barcode Scanner Power Scan	1	DATALOGIC
19	JN12-M01	Barcode Printer. BARCODE LABEL THERMAL PRINTER	1	CITIZEN
20	CF-200	Machine Fan. 200 MM Cabin Fan Blue Colour	1	REMI
21	SHUNT_I/P-0-5A,O/P-0-75mV	I/P-0-5A,O/P-0-75mV	3	RISHABH
22	SI-101	CONVERTER, I/P-0-75mVDC, O/P-0-10VDC AUX SUPPLY - 24...60V DC	3	RISHABH
23	15X40MM(HXL) ENGRAVING	NAME PLATE. SQUARE ENGRAVING		LOCAL
24	ROUND Dia 22.5MM	NAME PLATE. ROUND ENGRAVING		LOCAL
25	857-304	Slim Relay.Relay module; Nominal input voltage: 24 VDC; 1 changeover contact; Limiting continuous current: 6 A; Yellow status indicator; Module width: 6 mm; 2,50 mm ² ; gray	38	WAGO
26	2946-859-410-ND	SHORT LINK.PUSH-IN TYPE JUMPER BAR; 10-WAY	4	WAGO
27	KUDF 4AD	Fuse Terminal. Dinrail Mouniting 800V, 6.3A Fuse Terminal,4Sq.mm Wire Size	6	ELMEX
28	VCF1	MAINS ON/OFF.Mains ON/OFF 32A	1	SCHNEIDER
29	A9N2P32C	MCB.2 Pole 32 Amp	1	SCHNEIDER
30	A9N2P06C	MCB.2P,6AMP	1	SCHNEIDER
31	A9N2P16C	MCB.2 POLE MCB,16A	1	SCHNEIDER
32	XB7EV04MP	INDICATION LAMP.POWER ON	1	SCHNEIDER

		INDICATOR. RED LED 230VAC		
33	T5 LED	TUBELIGHT. 1 FT LED TUBE LIGHT	1	PHILIPS
34	3SE3-020-OA	LIMIT SWITCH.Siemens 1NO+1NC Contact Type Limit Switch 3SE3 020-0A	1	SIEMENS
35	3" Fan Filter Jigo	Fan Filter.3" Fan Filter	2	JIGO
36	3" Fan Grill	Fan Grill.3" Fan Grill	2	JIGO
37	3" Fan 230VAC	Fan.3" Fan 230VAC	2	JIGO
38				
39	NHP-24VDC-5A	SAMPS.SMPS 24VDC;5A	1	NHP
40	6755 62	SERVICE SOCKET.MYLINC PLATES -2 MODULES	1	LEGRAND
41	673302	SERVICE SOCKET.MYLINC SURFACE BOX -2 MODULES	1	LEGRAND
42	6755 01	SERVICE SOCKET MYLINC 6 AMP SWITCH SP	2	LEGRAND
43	6755 50	SERVICE SOCKET MYLINC 6 AMP SOCKET 2/3 PIN	1	LEGRAND
44	6733 04	SURFACE BOX.MYLINC SURFACE BOX- 4 MODULES	1	LEGRAND
45	6755 64	MODULE PLATE.MYLINC 4 MODULE PLATE-WHITE	1	LEGRAND
46	ZMPS15-10 With Display	Power Supply.DC Variable Power Supply 0 to 15VDC, 230VAC Input Supply	1	ZEAL
47	675974	MCB Switch.Mylinec Modular SP MCB 16A	2	Legrand

• **PNEUMATIC BOM:**

Project No.: A21082				
No	PART NAME	PART NO.	QTY	MAKE
1	SPRING BALNCER	HSB-3	1	ENDO
2	LINEAR BEARING	SC16UU	1	SAMIK
3	DRAW CHAIN		1	IGUS
4	REJECTION BIN	FPO CRATE	1	NILKAMAL
5	STRAIGHT FITTING	KQ2H-08-U02A	1	SMC
6	ON OFF VALVE	VHS20-02A	1	SMC
7	SILENSOR	AN10-01	1	SMC
8	BRACKET	Y200-A	2	SMC
9	FRL	AW20-02BCG1-B	1	SMC
10	L FITTING	KQ2L06-01A	1	SMC
11	T-INTERFACE	Y210-02-A	1	SMC
12	PRESSURE SWITCH	ISE20A-T-01-JD	1	SMC
13	SPACER WITH BRACKET BRACKET	Y200T-A	1	SMC
14	MODULAR ADAPTOR	E210-U02	1	SMC
15	3/2 VALVE	VT317-5DZ-02F	1	SMC
16	SILENSOR	AN15-02	1	SMC
17	L FITTING	KQ2L06-02A	1	SMC
18	DCV	SY5120-5LZ-01	2	SMC
19	DCV	SY5120-5LZ-01	2	SMC
20	STRAIGHT FITTING	KQ2H06-U01A	10	SMC
21	MANIFOLD	SS5Y5-20-06	1	SMC
22	STRAIGHT FITTING	KQ2H06-U02A	3	SMC
23	SILENSOR	AN15-02	3	SMC
24	PLUG	PL-02	3	SMC
25	BLANKING PLATE ASSEMBLY	SY5000-26-20A	2	SMC
26	KDM CONNECTOR	KDM10-06	1	SMC

7. FLOW CHART:



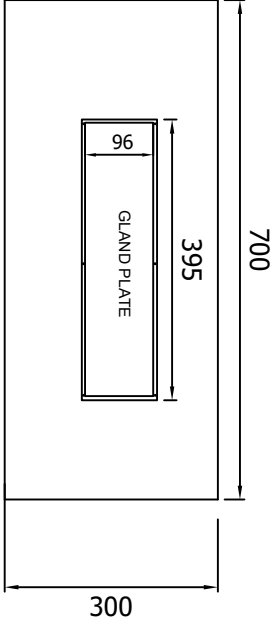
8. TROUBLESHOOT:

- **Emergency Button Pressed alarm:** Release the Emergency Button and Reset alarm from auto screen.
- **Air Pressure Alarm:** Check if air supply is in required range and reset the alarm from auto screen.
- **Clamp Reed Switch Alarm:** Reset alarm from auto screen. Check if the reed switch is glowing by operating the cylinder manually. If not, adjust the position of reed switch to sense the clamp position. Check if correct Reed Switch F/b is enabled in configuration.
- **Fixture selection is wrong alarm:** Check if the program selection is correct according to the model. Reset alarm from auto screen.
- **Glow Test Lamp NOK:** This alarm indicates that the current of a specific lamp is not in required range. Reset alarm from auto screen. Retest the lamp for confirmation, if the alarm occurs again, the Lamp is Not OK. Reset alarm from auto screen.
- **Part Presence Alarm:** this alarm occurs when a part is missing to start cycle. Reset alarm from auto screen. Check & place the part correctly, check if the sensor is sensing the part correctly. Check if the correct sensor is enabled in Configuration.
- **PCB Scan NOK:** This indicates that the scanned part is not correct. Reset alarm from auto screen. Scan the correct part. Check if correct part number is set in configuration.

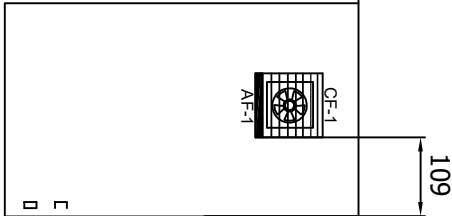
9. ELECTRICAL & PNEUMATIC DRAWING:

PANEL BOX-Its powder coated
SIZE:- 500(H) x 700(W) x 300(D) MM

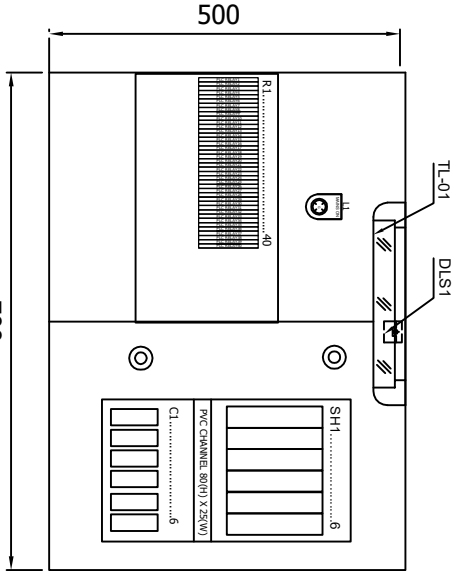
MOUNTING PLATE
SIZE:-450(H) x 650(W)



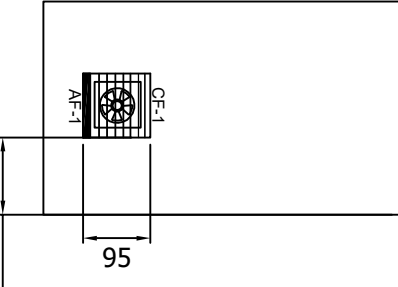
TOP VIEW



L.H.SIDE VIEW



FRONT VIEW



R.H.SIDE VIEW

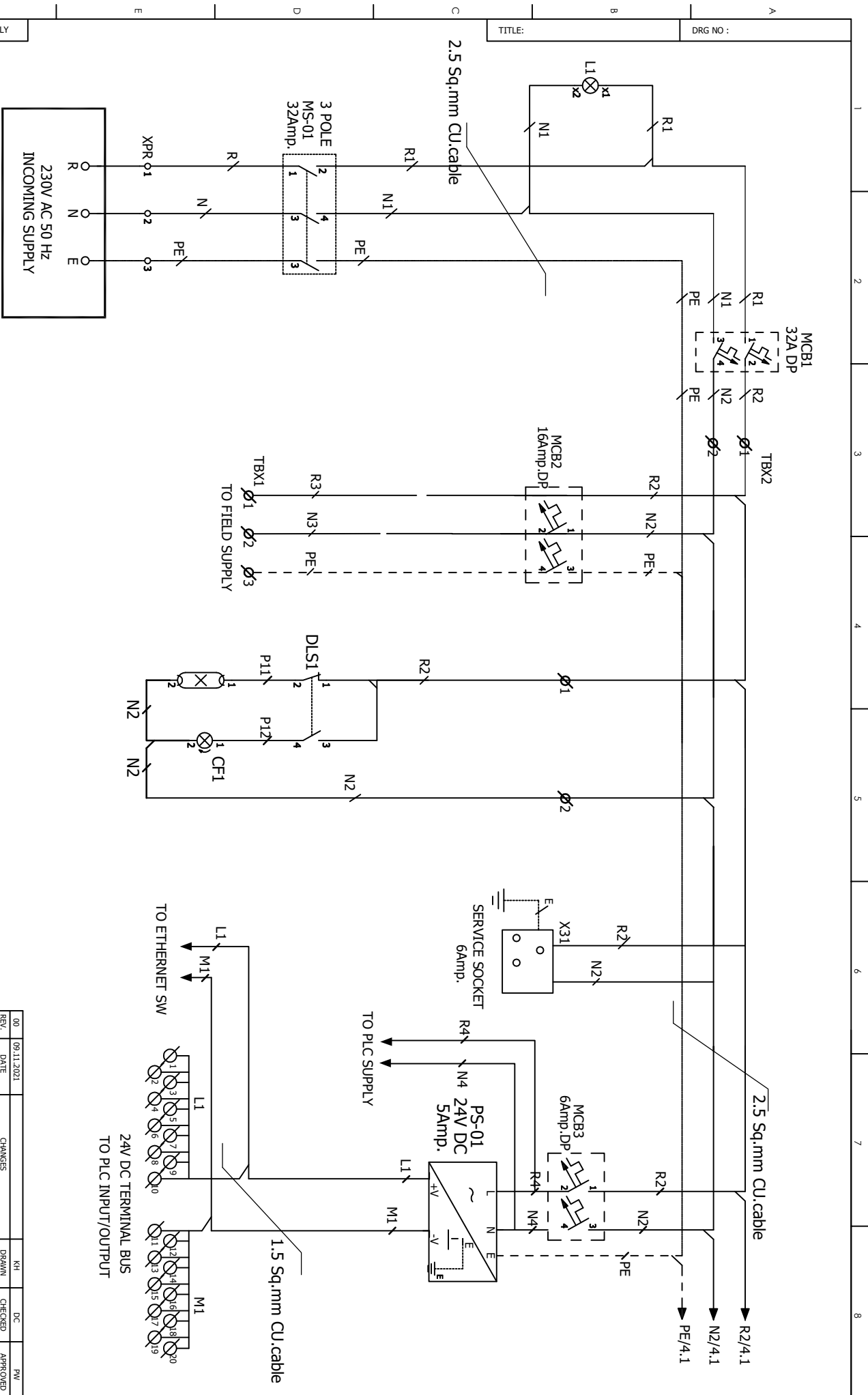
TITLE: DRG NO :

CHANGES VIA CAD ONLY		
WIRE SIZE & COLOUR DETAILS :-		
1. 230V AC	:- 2.5,1.5 SQ. MM RED	
2. 0V AC	:- 2.5,1.5 SQ. MM BLACK	
3. PE	:- 2.5,1.5 SQ. MM YELLOW/GREEN	
4. 24V DC	:- 0.5 SQ. MM BROWN	
5. 0V DC	:- 0.5 SQ. MM BLUE	
6. DIGITAL INPUT	:- 0.5 SQ. MM GRAY	
7. DIGITAL OUTPUT	:- 0.5 SQ. MM WHITE	

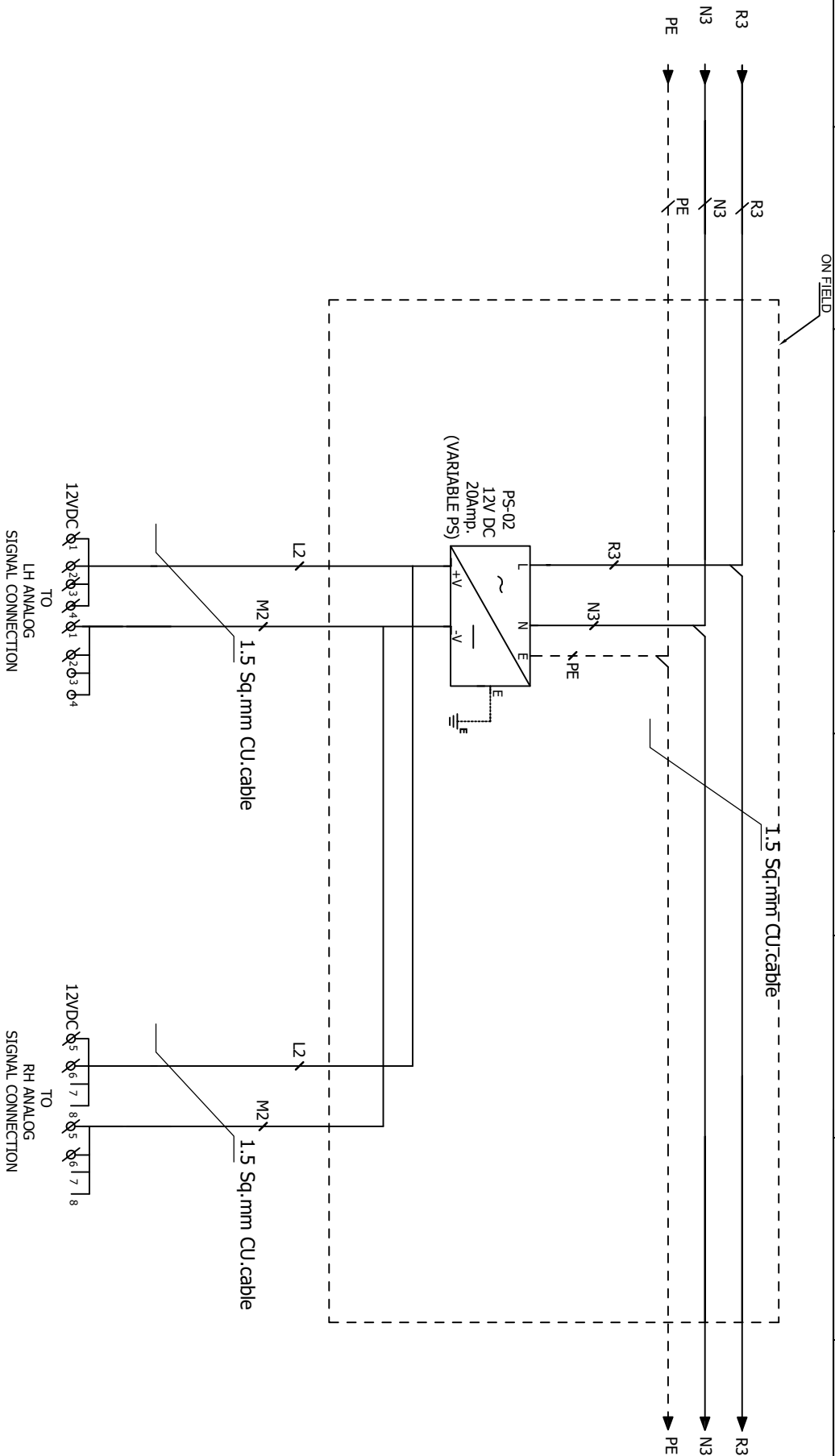
REV.	DATE	CHANGES	MB	DC	PNV
00	11.04.2022				
CLIENT:- MVM					
PROJECT:- A2002					
TITLE :-					
Housing to reflector screening assembly in (TFS Standard Truck)					
SCALE :- MTS					
QTY :- 01 No.					
SHEET 1 of 17					
DRG NO :- A21002					
REV :- 00					



INFINITY AUTOMATION SYSTEMS PVT. LTD.
WORKS NO.802A,NIMBA BANGLORE HIGHWAY/NEAR
JSPM ENGG. COLLEGE,ATMAWADE, PUNE - 411033



00	09.11.2024					
REV.	DATE	CHANGES	KH	DC	PW	
CLIENT:		DRAWN		CHECKED	APPROVED	
MM						
PROJECT:		 INFINITY automation systems pvt. ltd. WORE, Unit 2.5, No. 80/2, Mumbai Bangalore Bypass Highway Near ISPAT Institute, Tahsawade, Pune 411033				
NO. OF SHEETS: 5/14						
TITLE: -						
INCOMING POWER WIRING DETAIL DRAWING FOR RLC CONTROL PANEL.						
SCALE: - NTS	QTY: - 01 No.	SHEET: 3 of 17	DRG NO: -A21082	REV: - 00		



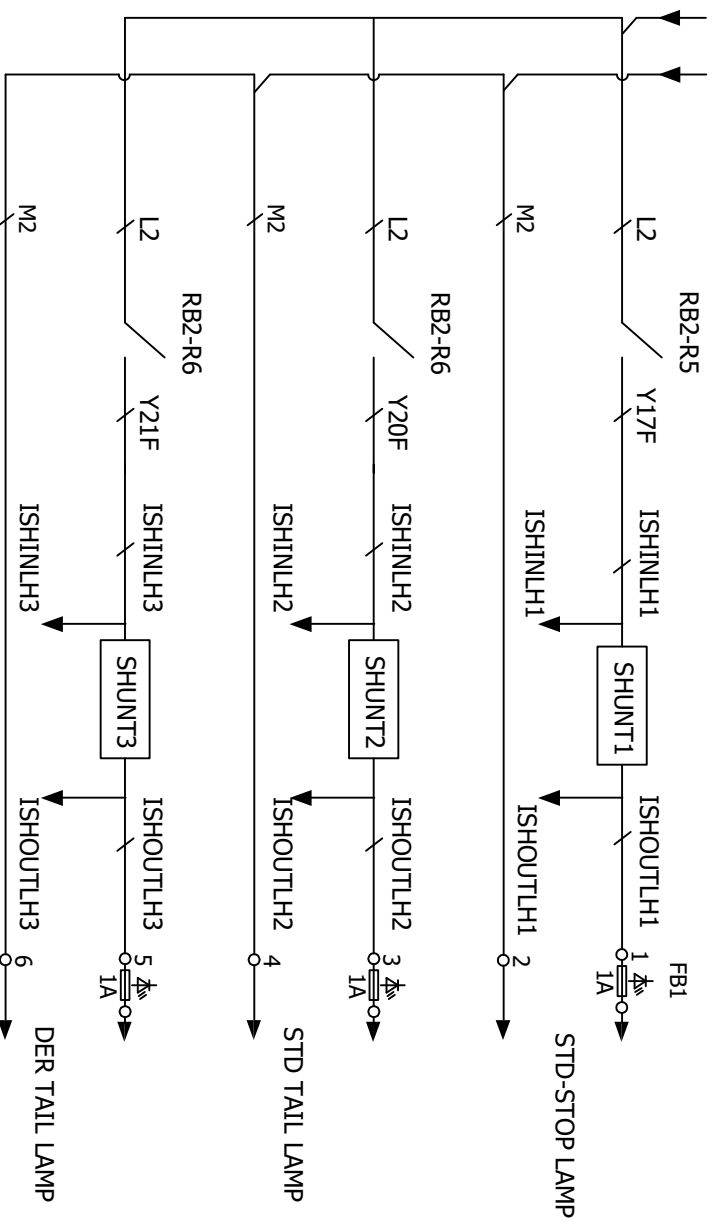
00	11.04.2022	CHANGES	KH	DC	PW
REV.	DATE	DRAWN	CHECKED	APPROVED	
CLIENT:-					
NAME					
PROJECT:-					
42302					
TITLE :-					
Housing to reflector screwing assembly LH (YFG Standard Trunk)					
SCALE : -NTS					
QTY:- 01 No.		SHEET: 4 of 17		DRG NO :- A21082	
REV:- 00		PROJECT		INFINITY	
				INFINITY AUTOMATION SYSTEMS PVT.LTD.	
				WORK: S NO.80/2A,MUMBAI BANGLORE HIGHWAY/NEAR	
				JSPM ENGS. COLLEGE, WATHANDE, PUNE - 411033	

FROM PS-02

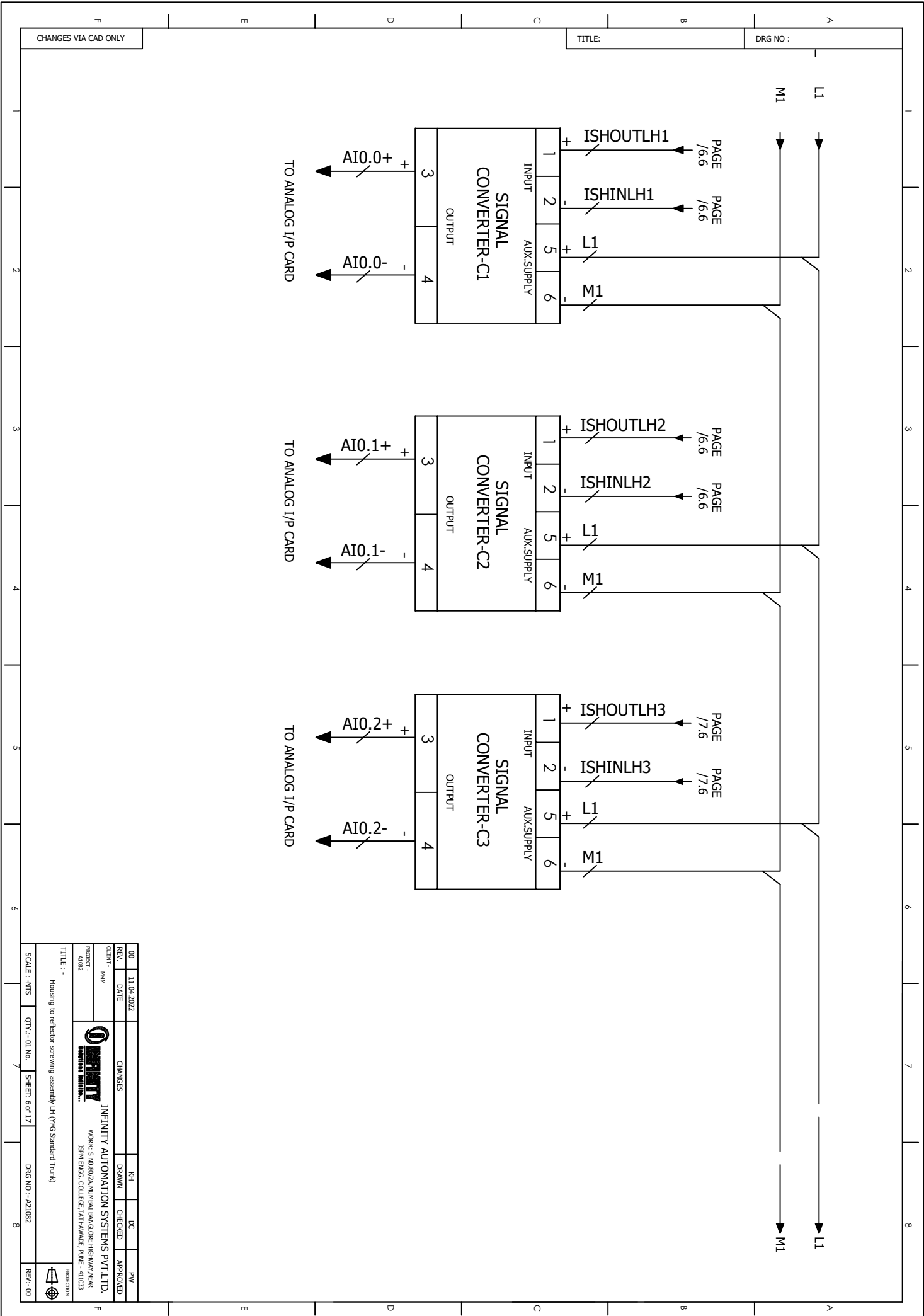
PAGE:5.4

L2 M2

LH ANALOG SIGNAL CONNECTION



00	11/04/2022						
REV.	DATE	CHANGES	DRAWN	CHECKED	APPROVED		
CLIENT:	NAME	 INFINITY AUTOMATION SYSTEMS PVT. LTD.					
WORKS : NO.80/2,MURBARI BALNOCHE, NGAWA,MYAR JAWI ENGINE, COLLEGE, INTHAWADE, PUNE - 411033							

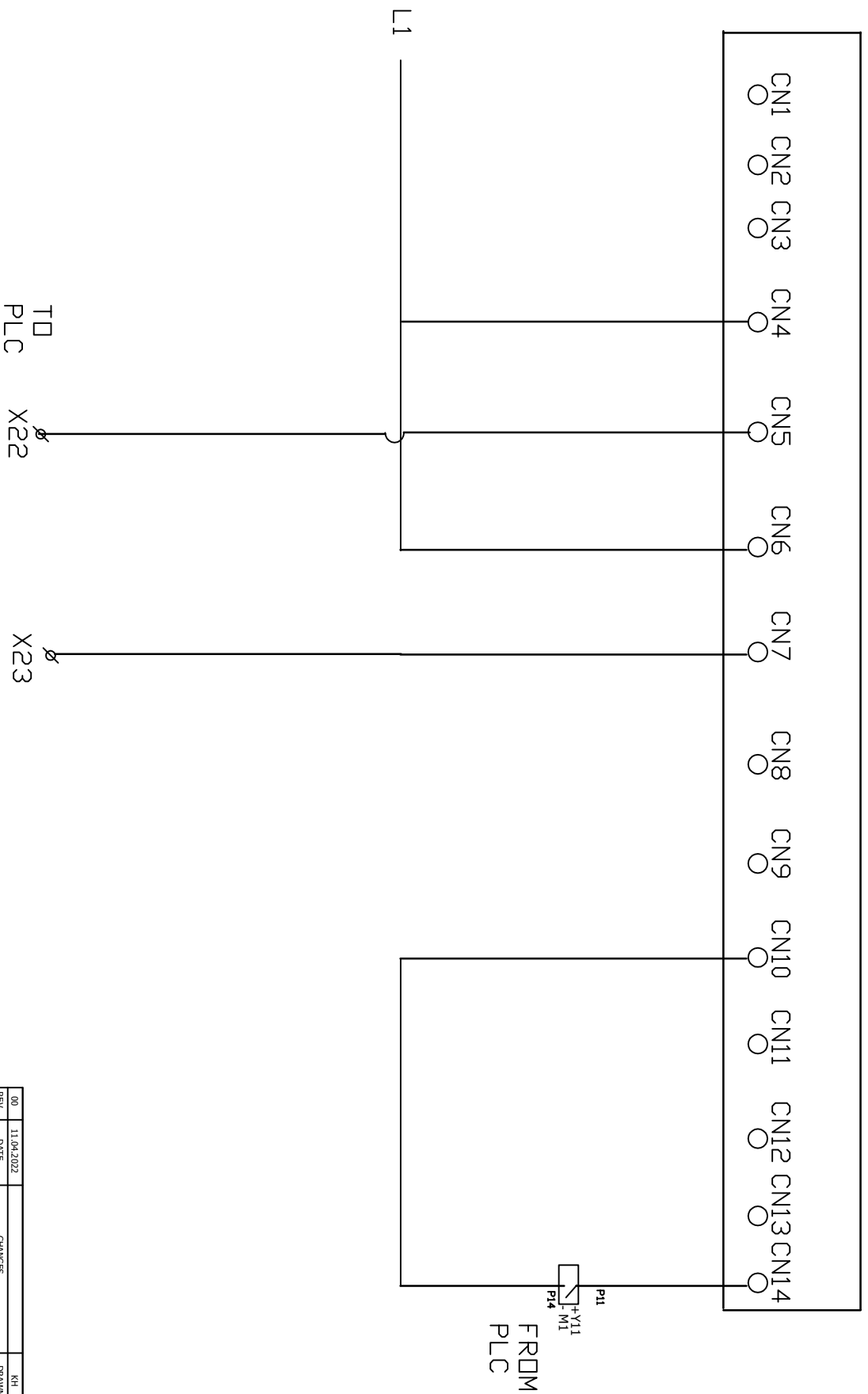


1 OF 1

24V DC
→ L1

Ip Spare

SCREW GUN LOGIC

[illegible]

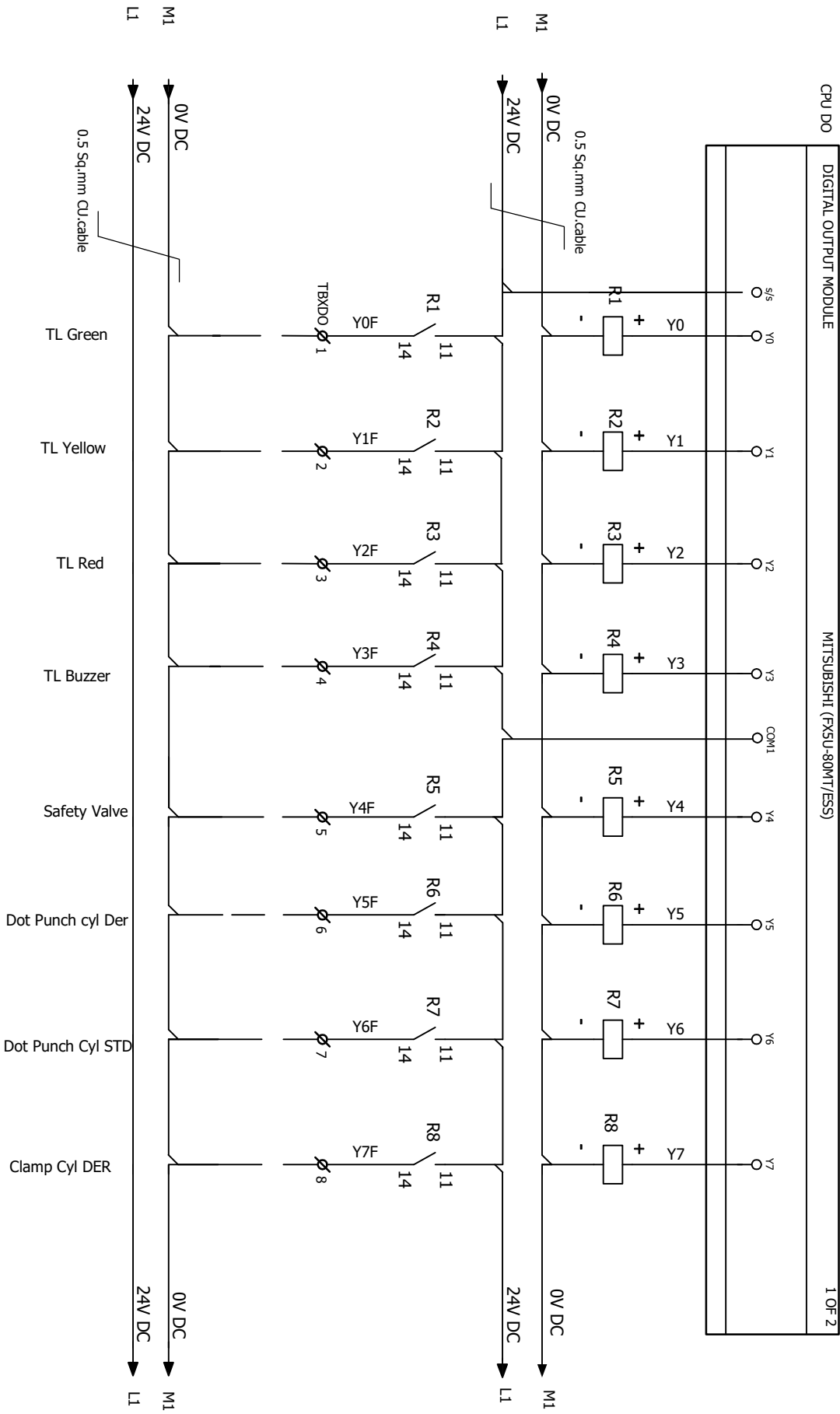
NOTE :-

IRAPAT
Soluções Inteligentes

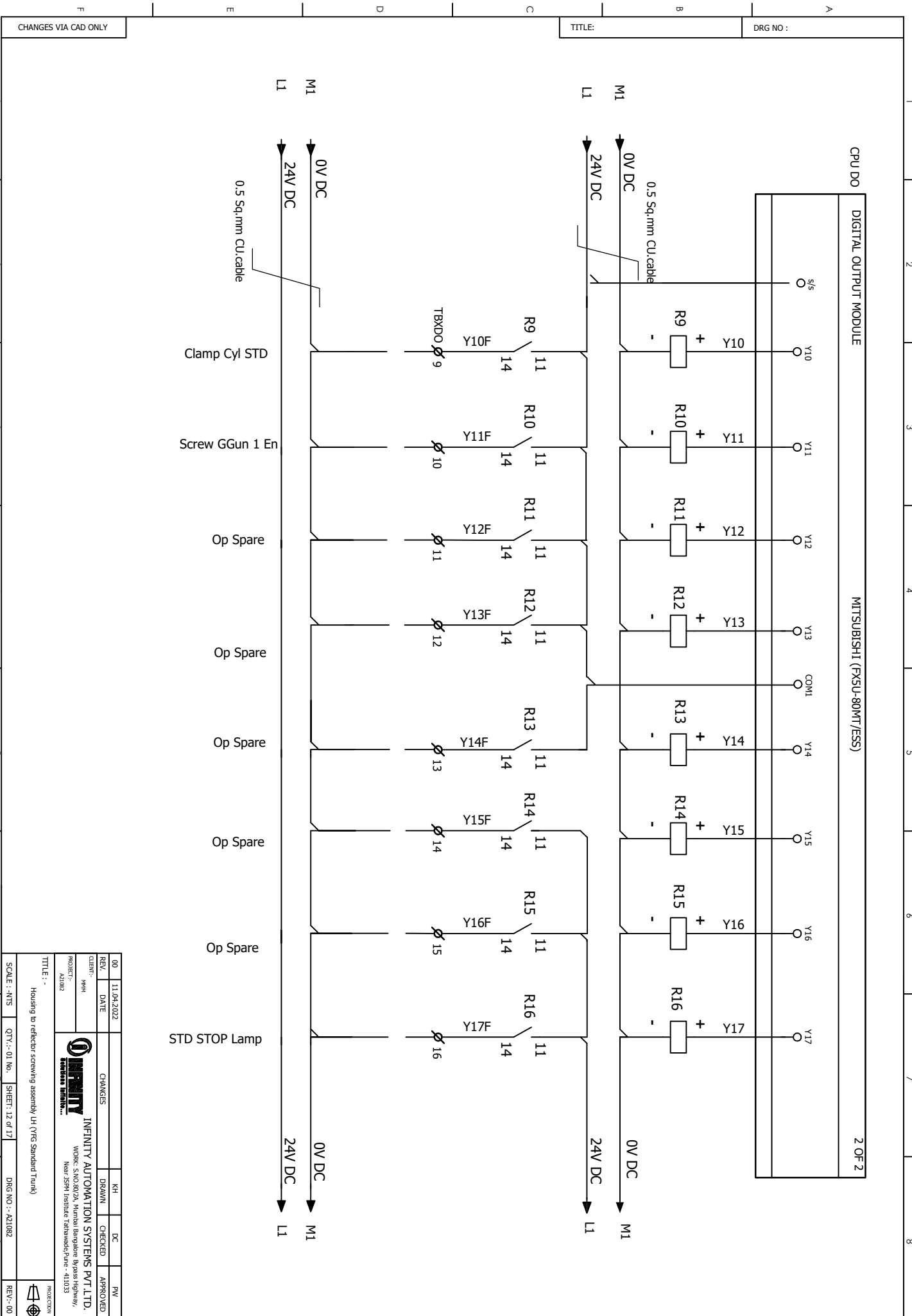
INFINITY AUTOMATION SYSTEMS PVT.LTD.
WORK: S.NO.80/2A,MUMBAI BANGLORE HIGHWAY,NEAR
JSPM ENGG. COLLEGE,TATHAWADE, PUNE - 411033

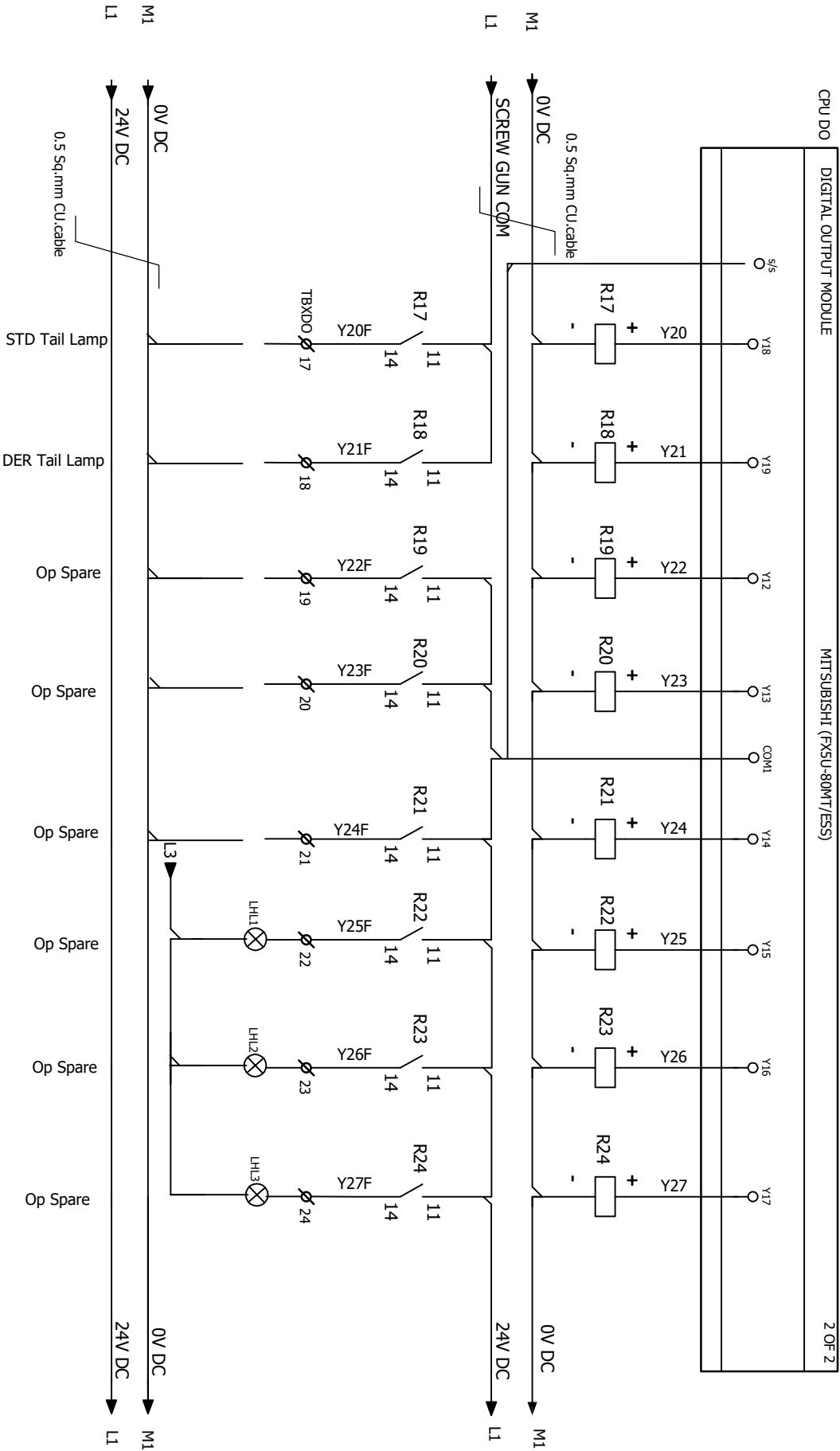
PROJECTION

SCALE : -NTS	QTY :- 01 No.	SHEET : 10 of 17	DRG NO :- A21082	REV:- 00
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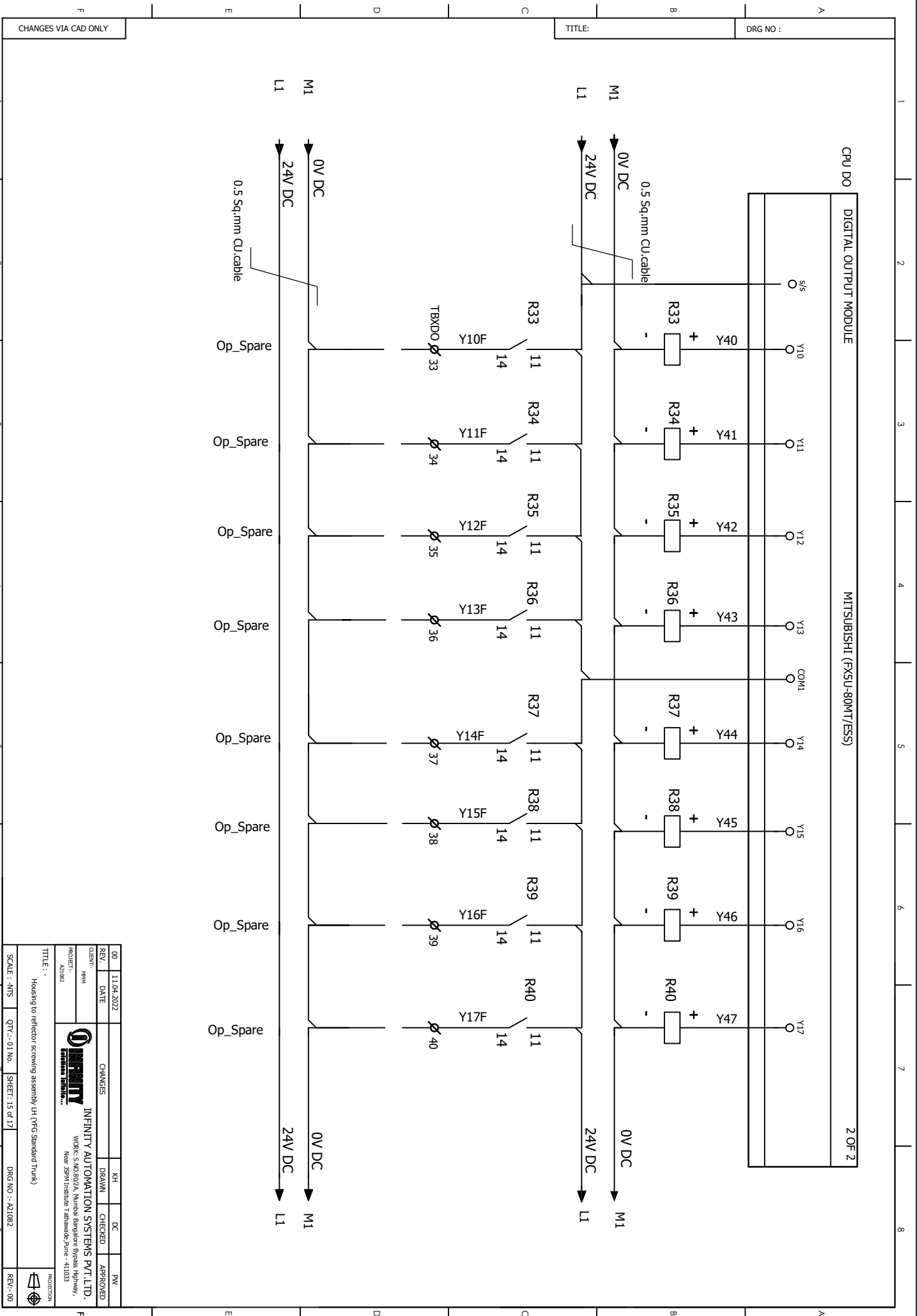


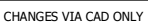
00	11.04.2022	CHANGES	KH	DC	PW
REV.	DATE		DRAWN	CHECKED	APPROVED
CLIENT:-	MMN	INFINITY AUTOMATION SYSTEMS PVT. LTD.			
PROJECT:-	A21082	WORK: S.NO.80/2A, Mumbai Bangalore Bypass Highway, Near 29th Institute Talavade,Pune - 411033			
TITLE :-				PROJECTION	
Housing to reflector screwing assembly LH (YFG Standard Trunk)					
SCALE : MTS	QTY :- 01 No.	SHEET: 11 of 17	DRG NO :- A21082	REV:- 00	





00	11.04.2022					
REV.	DATE	CHANGES	IN	DRAWN	CHECKED	APPROVED
CUSTOMER	DATE					
PROJECT						
A21082						
INFINITY AUTOMATION SYSTEMS PVT.LTD.						
WORK: S.NO.80/7A, Mumbai Bangalore Bypass Highway,						
Near JSRP Institute, Tathawade,Pune - 411033						
Tathawade, Pune						
TITLE :-						
Housing to reflector screwing assembly LH (YFG Standard Trunk)						
SCALE :- A:NTS	QTY :- 01 No.	SHEET: 13 of 17	DRG NO :- A21082	REV: 00		





00	11.04.2022						
REV.	DATE	CHANGES	KK	DC		PV	
CURT:							
9494							
PROJECT:							
A21082							
 INFINITY AUTOMATION SYSTEMS PVT. LTD. WORKS: S.NO.80/2A, Mumbai Bangalore Bypass Highway, Near JSRP Institute, Rahisarwad,Pune - 411033							
TITLE :-		HOUSING for reflector screwing assembly LH (VTS Standard Trunk)					
SCALE :- NTS	QTY:- 01 No.	SHEET: 16 of 17	Dwg NO :- A21082			REV:- 00	

TERMINALS DETAILS - TERMINALS DETAILS - POWER/CONTROL TERMINALS

XPR	
WIRE NO.	TERM. NO.
R	1
N	2
PE	3

TBXI	
WIRE NO.	TERM. NO.
R3	1
N3	2
PE	3

TBX2	
TERM. NO.	WIRE NO.
1	R2
2	N2
3	PE

24V & 0V DC TB1			
TERM. NO.	1	3	5
WIRE NO.	L1	L1	L1
TERM. NO.	2	4	6
WIRE NO.	M1	M1	M1
TERM. NO.	8		

TBXD1																					
	TERM. NO.	1	3	5	7	9	11	13	15	17	19	21	23	25	27	29	31	33	35	37	39
	WIRE NO.	X0	X2	X4	X6	X10	X12	X14	X16	X20	X22	X24	X26	X30	X32	X34	X36	X40	X42	X44	X46
		TERM. NO.	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38
	WIRE NO.	X1	X3	X5	X7	X11	X13	X15	X17	X21	X23	X25	X27	X31	X33	X35	X37	X41	X43	X45	X47

ITEM. NO.		WIRE NO.
1	3	
5	7	Y0F
9	11	Y2F
13	15	Y4F
17	19	Y6F
21	23	Y10F
25	27	Y12F
29	31	Y14F
33	35	Y16F
37	39	Y20F
		Y22F
		Y24F
		Y26F
		Y30F
		Y32F
		Y34F
		Y36F
		Y40F
		Y42F
		Y44F
		Y46F

00	11/04/2022				PW
REV.	DATE	CHANGES	RH	DC	APPROVED
CLIENT:	AMM		DRAWN	CHECKED	
PROJECT:	AZ1082				
 INFINITY AUTOMATION SYSTEMS PVT. LTD. WORKS : NO.80/2A,MARBAI,BANGLORE,HYDRABAD 2ND FLOOR, COLLEGE MAINROAD, PHONE : 410083					
TITLE :- Housing to reflector screwing assembly LH (YES Standard Trunk)					
SCALE :- NTS	QTY:- 01 No.	SHEET- 17 of 17	DRG NO :- AZ1082		REV.-00
					